DR. MARIA BEATRIZ WALTER COSTA

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SUMMARY

Bioinformatician (PhD) with 10+ years of experience in genomics, transcriptomics and large-scale biological data analysis. Proven track record in biomarker discovery, machine learning and statistical modeling across academic and clinical research environments. Experienced in collaborating with interdisciplinary teams and translating complex biological data into actionable insights. Skilled in developing scalable and reproducible bioinformatics workflows for high-throughput data analysis.

WORK EXPERIENCE

Friedrich-Schiller University Jena — Jena, Germany
Scientific Researcher – Bioinformatics (fixed-term) — *10/2022 - 09/2025*

- Analyzed large-scale microbial multi-omics data; identified 80 environmental biomarkers (machine learning)
- Designed **scalable and reproducible workflow** (Snakemake) for genomics data (extracts **150,000 features from 13,000 microbial genomes**), leveraging parallel computing on high-performance clusters
- Co-supervised a PhD student and contributed to teaching 3 graduate-level courses on NGS analysis

University of Leipzig Medical Center — Leipzig, Germany
Scientific Researcher – Bioinformatics (fixed-term) — *02/2020 - 12/2021*

- Applied statistical methods (R) to clinical data analysis (**260,000 patients, 20 million laboratory values, 3 AMPEL frameworks**) to support the automated diagnosis of 3 acute medical conditions
- Worked closely with physicians and industry partners to build AMPEL Clinical Decision Support System
- Structured a clinical knowledge data flow to improve downstream analysis (SAP-based data)

Brazilian Agricultural Research Corporation (EMBRAPA) — Brasilia/DF, Brazil
Consultant (fixed-term) — *11/2018 - 03/2019*

- Supported strain-engineering decisions by analyzing NGS data/RNA-seq (Illumina HiSeq) to find key genes to optimize ethanol production (R, DESeq2)

University of Leipzig — Leipzig, Germany
Scientific Researcher – Bioinformatics (fixed-term) — *04/2017 - 03/2018*

- Developed a **new statistical method**, `SSS-test` (*BMC Bioinformatics*), analyzed 87,000 structures for biomarker discovery and **identified 14 ncRNAs** with potential relevance to psychiatric disorders

TECHNICAL SKILLS

Programming: Python, R, Perl, Bash, SQL (MySQL)
Bioinformatics: genomics, transcriptomics, ncRNA analysis, statistical modeling, machine learning
Workflow & Infrastructure: Snakemake, Linux/Unix, HPC platforms (Slurm, SGE), Git
Portfolio: `https://github.com/waltercostamb`

EDUCATION

Dr. rer. nat./PhD in Computer Science (*magna cum laude*) Leipzig, Germany
University of Leipzig, Chair of Bioinformatics *2013 - 2018*

· Specialization in method development in Computational Biology (6 publications, 3 methods)

MSc in Molecular Biology Brasilia/DF, Brazil
University of Brasilia, Graduate Program of Molecular Biology *2010 - 2012*

· Specialization in NGS-based analysis in immunogenomics (1 publication)

BSc in Biological Sciences Brasilia/DF, Brazil
University of Brasilia, Institute of Biological Sciences *2006 - 2010*

· Specialization in Bioinformatics

LANGUAGES

English: Fluent (C2, Cambridge CPE)
Portuguese: Native (C2)
German: Advanced (C1)
Spanish: Advanced (B2/C1)

SELECTED PRODUCTS

Reproducible high-throughput workflow (Snakemake)
`FxTractor`

· Workflow for feature extraction from microbial genomes (150,000 features across 13,000 genomes)

`https://github.com/MGXlab/FxTractor`

Novel method for ncRNA analysis
`SSS-test`

· Bioinformatics method for ncRNA analysis integrating sequence and structural information

`https://github.com/waltercostamb/SSS-test`

REFERENCES

Univ.-Prof. Dr. med. Thorsten Kaiser
Chefarz/Institutsleiter, Klinikum Lippe GmbH, Universitätsklinikum OWL of the University Bielefeld

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